

AST1410 MESA Practical 3

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To generate isochrones, 100 stars were simulated in MESA with solar metallicity $Z = 0.0142$ based on Asplund et al., 2009, with uniformly log-spaced masses ranging from $0.1M_{\odot}$ to $50M_{\odot}$. Stopping conditions were chosen to detect the helium-burning phase, occurring a bit after main sequence turnoff; they were determined through trial and error over several runs of simulations at subintervals of the full mass range, by visual inspection of the resulting HR tracks. They are the following:

- Age 10 billion years to stop low mass simulations after a reasonable amount of time (in hindsight, a potential mistake for the purpose of estimating globular cluster ages with isochrones since they may be older than 10 Gyr, but it did not significantly affect results).
- Core temperature 10^8 K, approximately when helium ignites. Used for all masses from $0.1M_{\odot}$ to $1.2317M_{\odot}$.
- Core temperature $10^{7.5}$ K, for $1.3115M_{\odot}$ to $2.0351M_{\odot}$ inclusive. The motivation for this was similarly to detect helium ignition, but simulations began taking too long to run with the original 10^8 K limit despite visibly going off the main sequence, so the limit was simply reduced.
- Core temperature 10^8 K in addition to a relaxation of the upper limit on $\frac{dX}{X}$ for a single timestep to 1 for all remaining masses. The reasons for these changes were as follows. The temperature condition was revamped to 10^8 K since as mass increased, it appeared that simulations were ending prior to the star definitively departing the main sequence. This, however, caused simulations to run for very long again – it was found that the parameter constraining the timestep size was $\frac{dX}{X}$, indicative of the hydrogen supply of the star being depleted. Since this only occurred in the regime $X \sim 10^{-6}$, it was safe to relax the condition globally.

The HR tracks generated using the final set of simulation parameters are shown in Figure 1.

Isochrones were generated at the arbitrarily chosen range of 13 uniformly log-spaced times from 10 Myr to 10 Gyr. Since it was not determined how to force MESA to log global stellar parameters at an exact time value, the points in each isochrone were generated as follows. A relative time error of 20% was allowed for the selection of points from each star mass. Then, iterating over masses, the data logged at the point in time closest to the isochrone time, if within the 20% range, was added to the isochrone (if there was no such point, this star was skipped for this isochrone). This means the resulting isochrones themselves have an error in time of about 10%, but this is ignored for the rest of the analysis since it is much smaller over the relevant data ranges compared to other uncertainties.

Main sequence turnoff (MST) values for mass and luminosity were computed by averaging over the highest T_{eff} value on each isochrone (which naturally traces the characteristic rightward hook on the HR diagram) and any points within 0.02 dex or about 5% of that value. This error range and the isochrone time error range were iteratively chosen in tandem until each MST point set contained at least two points closely packed in this hook feature, making the MST slightly more robust (less sensitive to isochrone time error) and allowing uncertainty estimation using the standard deviation of the set of points averaged over. The isochrone plot, with these MST points highlighted in red, is shown in Figure 2.

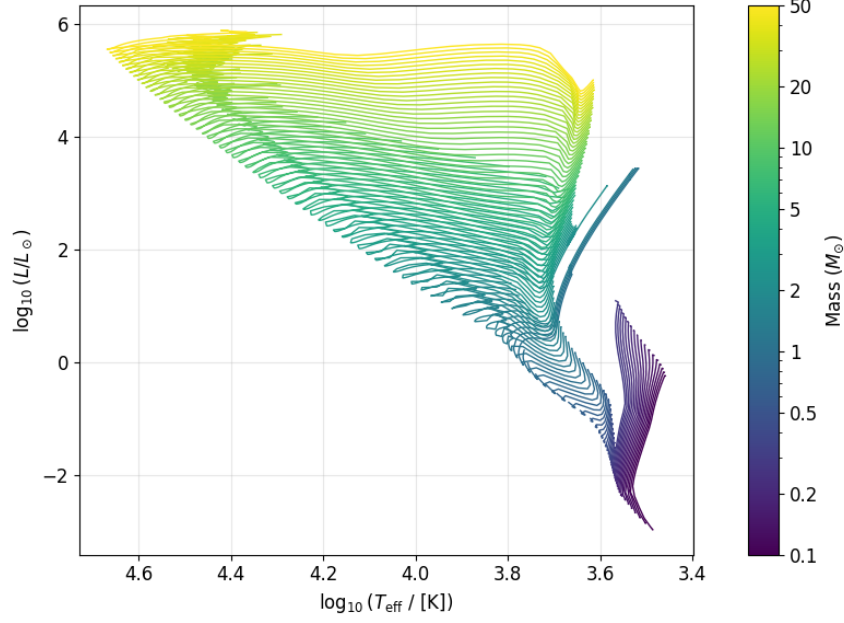


Figure 1: HR tracks of 100 solar metallicity stars ranging in mass from $0.1M_{\odot}$ to $50M_{\odot}$. For most stars (those of sufficient mass), there is a clear main sequence turnoff feature.

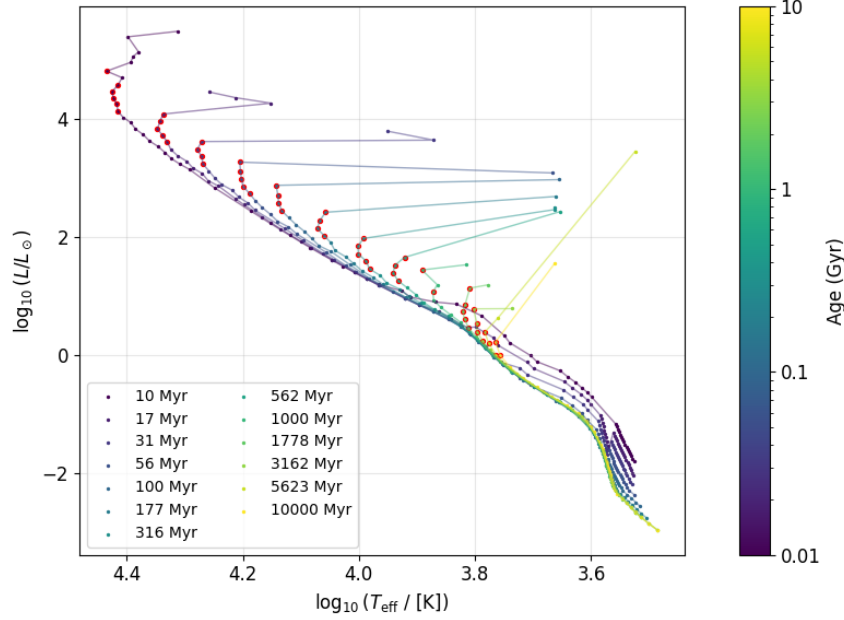


Figure 2: Simulated isochrones for solar metallicity stars, from 10 Myr to 10 Gyr. Some isochrones, especially younger ones, are more sensitive to the time error in choosing their points since they may probe pre- (or, for more massive stars, very post-) main sequence evolution, which is much more dynamic on short timescales compared to main sequence activity. Consequently, these curves may be more jagged or have unphysical features such as more massive giants having a higher T_{eff} at a given age. Points identified as being part of the MST feature for each isochrone are highlighted in red.

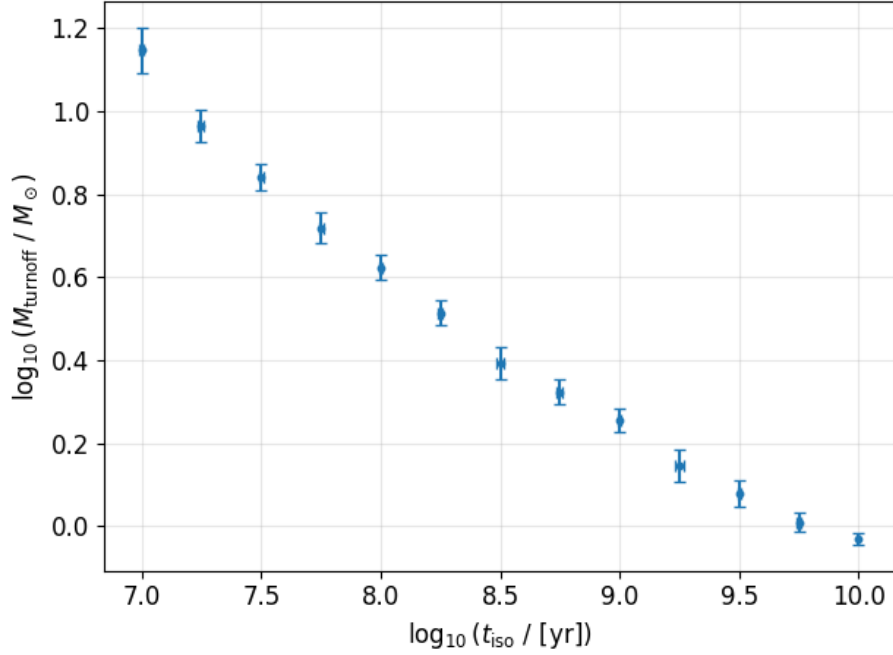


Figure 3: Simulated MST mass vs. isochrone age relation, plotted in log-log space so that power laws appear linear. The curve appears slightly non-linear.

With MST masses and times in hand, the relation between turnoff mass and age was plotted, with the result in Figure 3.

For the purpose of determining the age of Messier 3, we instead use a turnoff luminosity-time relation, since that is more directly applicable to the plot in Pols Figure 1.2 (a CMD). This relation was fit using `np.polyfit` with a linear function and inverse variance weighting to generate a slope m and offset b with uncertainties, including covariances. The relation and fit are shown in Figure 4. We first estimated the visual magnitude at turnoff from that figure; based on the shape of our simulated isochrones' MST regions, we take $V_{\text{turnoff}} = 19 \pm 1$ for the given CMD. We converted this to an absolute magnitude using a distance to M3 of $d_{\text{M3}} = 10400$ pc (from https://en.wikipedia.org/wiki/Messier_3), giving

$$M_V = V - 2 \cdot \left(100^{1/5}\right) \log_{10} \left(\frac{d_{\text{M3}}}{10\text{pc}} \right), \quad (1)$$

which has the same uncertainty as V , ignoring uncertainty in d_{M3} . This was subsequently converted to a log solar luminosity value using the bolometric magnitude of the sun $M_{V,\odot} = 4.74$ (based on Mamajek et al., 2015):

$$\log L := \log_{10} \left(\frac{L}{L_{\odot}} \right) = 100^{-1/5} \cdot (4.74 - M_V), \quad (2)$$

with $\sigma_{\log L} = 100^{-1/5} \cdot \sigma_{M_V}$. A cluster age was estimated by inverting the $\log L$ - $\log t$ linear fit and substituting the calculated value for $\log L$, i.e. $\log t_{\text{M3}} \approx \frac{\log L_{\text{M3}} - b}{m}$. Uncertainty in the yielded age estimate were computed using the standard partial derivative uncertainty propagation formula, including covariance terms:

$$\sigma_{\log t_{\text{M3}}} = \frac{1}{|m|} \sqrt{\sigma_{\log L_{\text{M3}}}^2 + \sigma_b^2 + (\log t_{\text{M3}})^2 \cdot \sigma_m^2 + 2(\log t_{\text{M3}}) \text{Cov}(m, b)}. \quad (3)$$

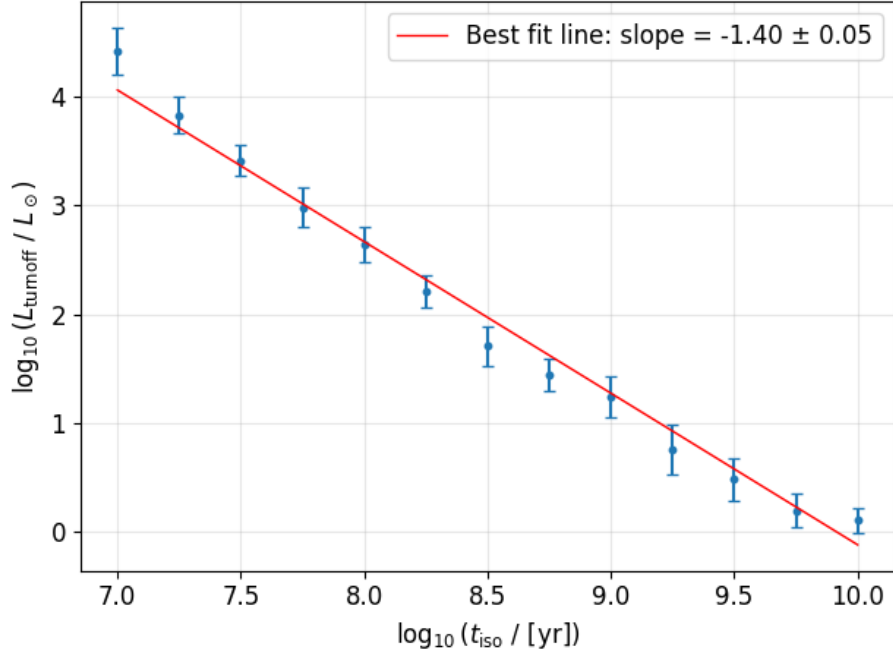


Figure 4: Simulated MST luminosity vs. isochrone age relation, plotted in log-log space so that power laws appear linear. The curve appears slightly non-linear, but potentially less so than the mass-time relation.

The resulting time estimate (appropriately converting log to linear uncertainty) is

$$t_{\text{M3}} = 4.54 \pm 3.03 \text{ Gyr} . \quad (4)$$

The large uncertainty is mostly due to the uncertainty in apparent magnitude from the turnoff point estimate of the actual M3 CMD. While correct in order of magnitude, this is less than half of the accepted age of the cluster of 11.39 Gyr (Forbes and Bridges, 2010). This large discrepancy is likely due in part to the systematic error in isochrone age associated with the metallicity chosen for our simulations - a typical globular cluster like M3 has much lower metallicity than the Sun.

Acknowledgements

My computer worked about as long as I did on this practical.

References

- Asplund, Martin et al. (Sept. 2009). “The chemical composition of the Sun”. en. In: *Annual Review of Astronomy and Astrophysics* 47.1. arXiv:0909.0948 [astro-ph], pp. 481–522. ISSN: 0066-4146, 1545-4282. DOI: 10.1146/annurev.astro.46.060407.145222. URL: <http://arxiv.org/abs/0909.0948> (visited on 02/15/2026).
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